

• LIVE DEVELOPMENT

EARLY ACCESS

URP Connector is being built.

A Unity Editor tool that connects, configures and upgrades your entire URP pipeline in one window — quality tiers, post-processing profiles, materials, terrain, shaders and more. No manual digging through Project Settings.

Want to follow progress or shape what gets built next? emaceart@gmail.com

v 1.3

CURRENT BUILD

EMACE ART

QUICK SETUP GUIDE — HOW TO CONNECT URP

Follow these steps in order. Each step takes under a minute.

#	Step	Where	Notes
1	Open URP Connector	Tools → EmacE → URP Connector	If URP is not installed the panel offers a one-click install button.
2	Select pipeline asset	Section 1 — Graphics Settings	Pick the asset matching your target quality (e.g. <i>HighQuality</i>). Use Auto-match by name to fill all quality levels automatically.
3	Create Very High asset	Section 1 — Quality Settings	Click <i>Ensure Very High Pipeline Asset</i> if the button is visible. Sets optimised shadow distance, cascades and LUT size automatically.
4	Click CONNECT URP	Section 1 — bottom	Assigns the pipeline asset to Graphics Settings and all enabled quality levels. Activates the <i>High</i> level. GUIDs are regenerated if the option is checked.
5	Upgrade materials	Section 2 — Materials	Assign the materials folder and click <i>Upgrade Materials to URP</i> . Unity's built-in converter remaps Standard → URP/Lit shaders.
6	Fix terrain	Section 3 — Terrain	Click <i>Fix Terrain</i> . Creates a URP/Terrain/Lit material per Terrain and assigns it. Save the scene afterwards.
7	Connect post-processing	Section 4 — Post Processing	Assign a profile folder. Select an existing profile and click <i>Connect</i> , or choose a climate preset and click <i>Generate</i> . A Global Volume is created if the scene has none. The camera is fixed automatically (Post Processing ON, FXAA).
8	Add SSAO	Section 5 — Renderer Features	Select the renderer asset and click <i>Add Screen Space Ambient Occlusion</i> . Defaults: Blue Noise, Intensity 3, Depth Normals, High quality.
9	Save the scene	File → Save (Ctrl+S)	The Global Volume and camera changes are scene objects — save to keep them.

Health check: The five status dots at the top of the panel (URP · Assets · Volume · Camera PP · SSAO) turn green as each step is completed. All green = fully connected.

MANUAL SETUP — HOW TO CONNECT URP WITHOUT THE TOOL

WIP

⚠ Work in Progress — An automated in-tool version of this walkthrough is under development. The steps below are the full manual fallback. Have suggestions or want to know when it ships? emaceart@gmail.com

STEP 1 — INSTALL UNIVERSAL RENDER PIPELINE PACKAGE

Action	Details
Open Package Manager	Window → Package Manager
Switch registry	Top-left dropdown → Unity Registry
Find & install	Search Universal RP → click Install . Wait for compilation.
Or via manifest.json	Add "com.unity.render-pipelines.universal": "14.x.x" to Packages/manifest.json dependencies.

STEP 2 — CREATE PIPELINE ASSET & RENDERER

Action	Details
Create URP Asset	Assets → Create → Rendering → URP Asset (with Universal Renderer) — Unity creates both the pipeline asset and a renderer asset.
Recommended location	Save inside a <i>Settings/</i> or <i>Rendering/</i> folder (e.g. Assets/Settings/).
Multiple quality tiers	Repeat for each level and add a suffix: -LowQuality, -MediumQuality, -HighQuality, -VeryHighQuality.

STEP 3 — ASSIGN TO GRAPHICS SETTINGS

Action	Details
Open Graphics Settings	Edit → Project Settings → Graphics
Assign pipeline asset	Drag the asset into Scriptable Render Pipeline Settings .
Verify	Scene view header switches to "Universal". No pink error shaders.

STEP 4 — WIRE PER-QUALITY-LEVEL ASSETS

Action	Details
Open Quality Settings	Edit → Project Settings → Quality
Per level	Click a quality level → find Render Pipeline Asset in the right panel → assign the matching tier asset.
Set default	Click ✓ next to the level you want active by default (e.g. <i>High</i>).

STEP 5 — CONVERT MATERIALS

Action	Details
Select materials	Select folder in Project window → Edit → Select All .
Run converter	Edit → Rendering → Materials → Convert Selected Built-in Materials to URP
Vertex Color materials	Use <i>Connect Vertex Color Shader</i> in URP Connector — shader is created automatically.
Review	Pink materials after conversion need manual shader assignment.

STEP 6 — FIX TERRAIN MATERIALS

Action	Details
Problem	Built-in terrain shaders are incompatible with URP — terrain appears pink or black.

Action	Details
Manual fix	Create a new material → shader Universal Render Pipeline / Terrain / Lit → assign to Terrain in Inspector → <i>Terrain Settings</i> → <i>Material</i> .
Automated	Use the <i>Fix Terrain</i> button in URP Connector.

STEP 7 — POST-PROCESSING VOLUME

Action	Details
Create Volume	Hierarchy: right-click → Volume → Global Volume .
Assign profile	Inspector → click New or assign an existing Volume Profile asset.
Enable on camera	Camera → Inspector → Rendering → Post Processing must be On .
Check renderer	URP Renderer asset → Post Processing enabled in Filtering.

STEP 8 — VERIFY

Check	Expected result
Scene view	No pink materials, no console errors about missing pipeline asset.
Game view	Post-processing effects visible (bloom, vignette, tonemapping).
Quality Settings	Each level has a dedicated URP asset assigned.
Shadows	Directional light casts shadows — check <i>Shadow Max Distance</i> on pipeline asset.

1 — RENDER PIPELINE

Quality Level	Pipeline Asset	Status
Fastest	New Universal Render Pipeline Asset-LowestQuality	active
Fast	New Universal Render Pipeline Asset-LowQuality	active
Simple	New Universal Render Pipeline Asset-LowQuality	active
Good	New Universal Render Pipeline Asset-MediumQuality	active
Beautiful	New Universal Render Pipeline Asset-HighQuality	active
Fantastic	New Universal Render Pipeline Asset-VeryHighQuality	active

VERY HIGH — SHADOW CONFIGURATION

Parameter	Value
Shadow Max Distance	160 m
Cascade Count	4
Depth Bias	0,00
Normal Bias	0,00
Soft Shadows	On
Color Grading Mode	LDR
LUT Size	32

2 — RENDERER FEATURES

SCREEN SPACE AMBIENT OCCLUSION (SSAO)

Renderer: **New Universal Render Pipeline Asset_Renderer**

Parameter	Value
Method	—
Intensity	0,16
Radius	0,33
Falloff	200
Direct Lighting	0,225
Source	Depth Normals
Blur Quality	High
Samples	Low
Downsample	Off
After Opaque	On

3 — POST-PROCESSING PROFILES

All profiles use Tonemapping (ACES) as base. Components: ColorAdjustments · Bloom · Vignette · WhiteBalance.

EA_19Th_PP_Profile Realistic				EA_19Th_PP_Profile			
Tonemapping		ACES		Tonemapping		ACES	
Contrast / Saturation		-15,4 / -40,2		Contrast / Saturation		-10 / -11	
Post Exposure		+0,4 EV		Post Exposure		+0,24 EV	
Bloom i / thr / sc		0,81 / 1,00 / 0,72		Bloom i / thr / sc		1,00 / 1,00 / 0,72	
Vignette i / sm		0,26 / 0,40		Vignette i / sm		0,26 / 0,40	
White Balance temp / tint		0° / 0		White Balance temp / tint		0° / 0	

Currently connected to scene Global Volume: **EA_19Th_PP_Profile Realistic**

4 — MATERIALS & TERRAIN

Asset type	Before	After
Scene materials	Standard (Built-in)	Universal Render Pipeline / Lit
Terrain material	Default-Terrain-Diffuse	Universal Render Pipeline / Terrain / Lit
Terrain material asset	—	{TerrainName}_URP_Terrain.mat — saved alongside TerrainData

Texture maps (Albedo, Normal, Metallic, Occlusion) remapped by Unity's built-in converter. Verify visually in Scene and Game view.

5 — ASSET INTEGRITY

All pipeline asset .meta files were assigned fresh GUIDs. References in .prefab .mat .unity .asset .controller .anim files across the project were updated to match. The pipeline folder is safe for redistribution and cross-project import without GUID collisions.

6 — VERTEX COLOR MATERIALS

Vertex Color is a technique where colour information is stored directly in the mesh's vertex data instead of a texture. It's commonly used in low-poly, stylised and performance-critical assets — zero texture memory, instant rendering, no UV unwrapping needed.

WHAT IT DOES

Feature	Description
Auto-detect	Scans the selected materials folder for any material whose shader name contains <i>VertexColor</i> or <i>VertColor</i> .
One-click conversion	Swaps the shader to Custom/URP/VertexColor and transfers <code>_Color</code> → <code>_BaseColor</code> so the tint is preserved.
Auto shader generation	If the URP vertex color shader is missing from the project it is created automatically at <code>Assets/EmacEArt/VertexColorURP.shader</code> — no manual file copying needed.
Lighting support	The generated shader uses PBR lighting (shadows, ambient, fog) — not flat/unlit. Matches the visual style of URP/Lit materials.
Color tint	A <i>Base Color</i> tint property multiplies over vertex colours — useful for global brightness or colour shifts per material.

REQUIREMENTS — WHEN IT WORKS

Requirement	Details
URP installed	The shader uses URP-specific HLSL includes. It will not compile in a Built-in or HDRP project.
Mesh has vertex colours	The mesh must have vertex colour data baked in (exported from Blender, Maya etc. with vertex colours enabled). If the mesh has no colour data Unity passes white (1,1,1,1) — result looks correct but flat.
Materials folder selected	The <i>Materials Folder</i> field in the URP Connector window must point to the folder containing the target materials.
Shader name convention	Source materials must use a shader whose name contains <i>VertexColor</i> or <i>VertColor</i> (case-insensitive). Materials already on Custom/URP/VertexColor are skipped.

WHEN IT WILL NOT WORK

Situation	Why
Mesh uses texture-based colour	Switching to the vertex color shader will drop the texture — albedo becomes black/white.
Shader name doesn't match convention	Materials with unrecognised shader names are skipped. Rename the shader or assign manually.
No materials folder set	The button is disabled until a folder is selected in the Materials section.
Built-in / HDRP pipeline	The Custom/URP/VertexColor shader requires URP packages to compile.

Tip: After conversion verify in the Scene view that colours look correct. If a material appears too dark, check that the mesh normals are correct — the shader uses PBR ambient (SampleSH) which responds to normal direction. If a material appears white/grey the mesh likely has no vertex colour data baked in.